



SthiraRoute

Sthira (स्थिर) — steady. A fleet plan that survives the day.

Problem Statement ID –

2

Problem Statement Title-

Smart Fleet Coordination and Logistics Management Platform

Theme-

Transportation & Logistics

PS Category-

Software

Team ID-

Team Name (Registered on portal)

Hack Shashtra

One board plans the shift, proves the load physically fits, and keeps the plan alive when the day breaks.

Measured on four city packs — Bengaluru, Mumbai, Delhi NCR and Hyderabad — over real OpenStreetMap road distances.

Route optimisation

3D load planning

Live tracking & re-planning



One board runs the whole shift

Plan every order onto the fleet · prove the load fits · dispatch to four desks · absorb breakdowns, rush orders and curfews — with every trade-off priced in rupees and the dispatcher always in charge.

Four failures in the problem statement, four mechanisms · red is the problem, green is what we built

Vehicles run half empty

Weight, volume *and* deck geometry per truck, plus ₹800 for every truck used — so the fleet consolidates itself.

Bengaluru: 15 orders on 2 trucks, 92% payload used

Routes planned by hand

OR-Tools over real OSRM road matrices, with delivery windows, curfews, cold-chain and two depots all binding the model.

10–31% less road distance, and legal to drive

No live visibility

Trucks move along the very polyline the map draws, with per-stop ETA, proof of delivery and a widening ring when GPS drops.

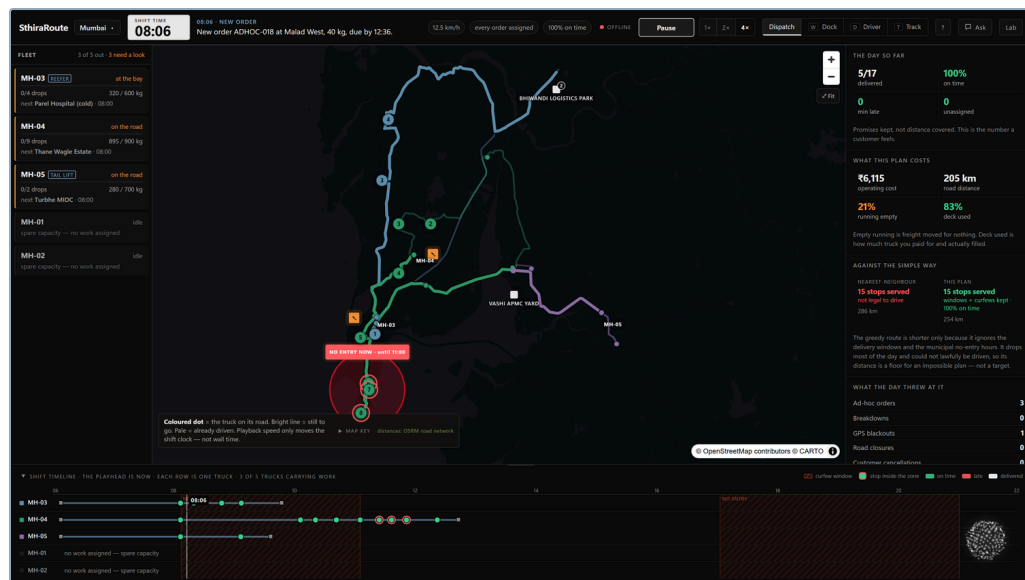
20 GPS blackouts survived, no freight lost

Everyone on a different page

One committed plan read from four desks — Dispatch, Dock, Driver, Track — plus an assistant that explains it in plain words.

Same plan object, four views, one source of truth

Working prototype · Mumbai, 08:06



Why this is not another routing demo

Churn has a price, so plans stay steady

Moving freight off the truck it was promised to costs ₹150 *inside the objective* — a re-plan is only offered when the gain beats the disruption.

Every route is checked against physics

A 3D packer proves the deck loadable before a truck moves: LIFO order, nothing floating, nothing stacked on fragile, centre of gravity in band.

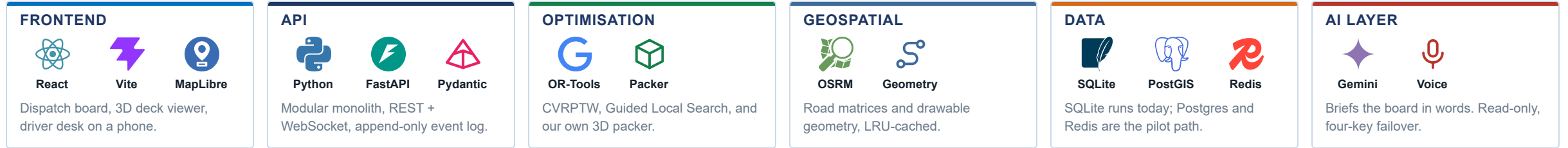
Routes obey the city, by construction

A no-entry window is time *and* geography: arrival times inside a banned zone are removed from the model, so the truck waits it out.

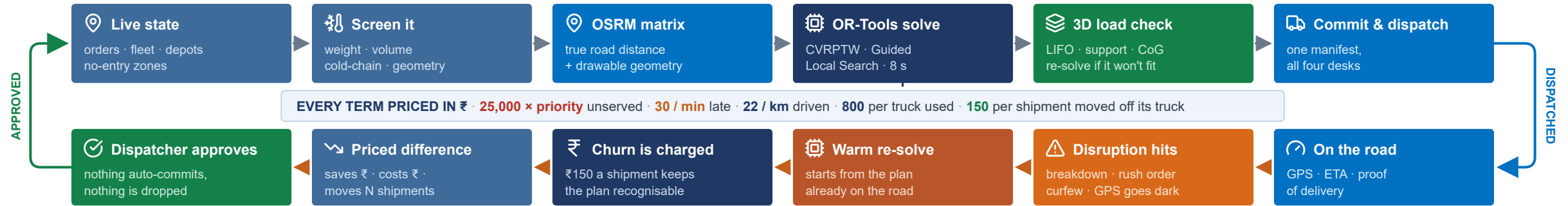
It never just answers INFEASIBLE

Every order carries a drop penalty, so a plan always returns — with a diagnosed reason and a relaxation menu for anything it could not serve.

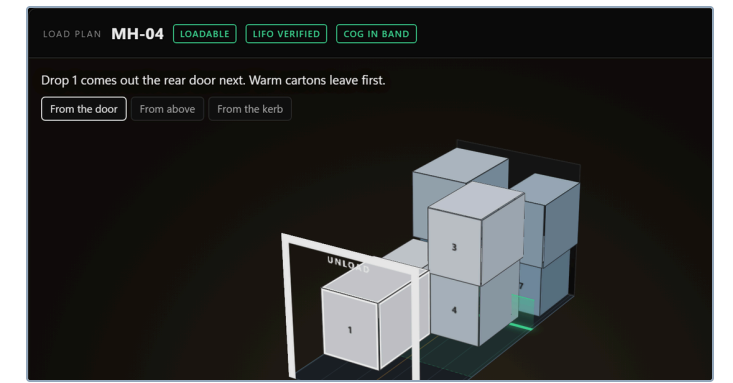
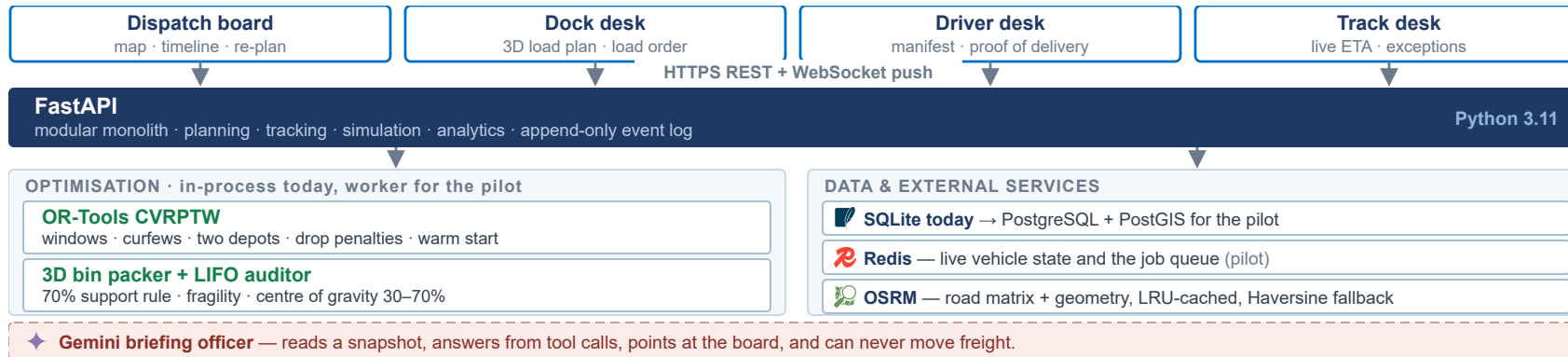
Technology stack · open source end to end — no paid routing key, no GPU, no licence



How a shift runs · plan across the top, recover along the bottom — one closed loop



System architecture · four desks, one API, one committed plan — and the load proved before a truck moves



FEASIBILITY AND VIABILITY

191

automated checks pass

53 pytest tests in 70 s · 93 interface checks in headless Chromium · 45 end-to-end assertions

10–31%

less road distance

measured in all four cities against a greedy baseline on the same OSRM roads

4

city packs, not one skin

each with its own depots, curfew windows, hazard corridors and congestion curve

₹0

licence cost to run

OR-Tools, OSRM, MapLibre, FastAPI, PostgreSQL — open source throughout

The growth path is a config file, not a rewrite · the solver never changes



Today — one laptop

BUILT AND MEASURED

SQLite, the solver in-process, the public OSRM. No Docker, no cluster, no map key. A whole shift plans in an 8-second solver budget.



Pilot — one VM

ONE DEPOT, ONE CITY, FOUR WEEKS

PostgreSQL + PostGIS and a self-hosted OSRM for one state. Runs in shadow beside the whiteboard while km and on-time rate are compared daily.



Scale — add a worker

CPU ONLY, NO PER-REQUEST FEE

Redis-backed workers take the solve off the request path. The recurring bill stays one server and one database — which is what mid-size fleets can afford.

Risks we can name, and what we already do about them



The public OSRM server is shared and rate-limited

Matrices and geometry are LRU-cached, and the system degrades to a Haversine fallback at a ×1.4 detour factor with the source labelled in the UI. A pilot self-hosts OSRM.



OpenStreetMap truck restrictions are thin in India

OSRM stays for geometry and matrices; Mappls vehicle-class routing supplies restriction-accurate distances the moment a pilot needs them. The solver does not care which provider answers.



A live re-plan cannot take minutes

Re-planning warm-starts from the plan already on the road and runs inside a hard time budget — four seconds for a disruption. A ₹150 churn price stops it reshuffling the day.



Curfew windows here are templates, not gazetted law

Every zone is per-city configuration meant to be sourced from traffic-police notification, and every seeded coordinate and window is labelled an assumption in the source.



Dispatchers do not trust a box that reshuffles their day

Recommend-and-approve: a re-plan lands as a proposal, drawn dashed on the map, showing what it saves, what it costs and how much freight it moves before anyone clicks.



A language model could invent numbers, or move freight

Gemini is read-only over a compact snapshot and answers from tool calls. It can point at the board and explain it; there is no code path by which it can alter a plan.

MEASURED, NOT PROJECTED

One Mumbai shift · 15 orders · 3 trucks
used of 5 · real OSRM road distances

254 km

road distance planned
greedy baseline: 286 km

100%

on time, 0 unassigned
every order served

18.4%

driven empty, exposed
the number nobody measures

68%

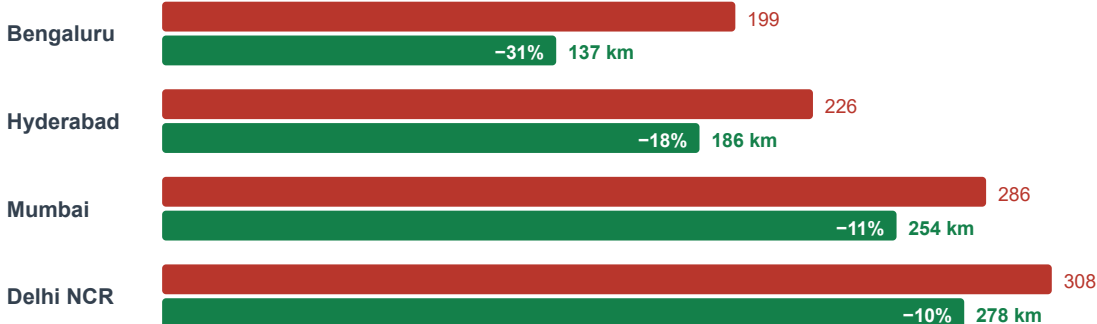
payload capacity used
weighed, not guessed

₹7,992

shift cost, itemised
every term in rupees

Distance, city by city · same orders, same roads

- greedy nearest-neighbour, ignores windows and curfews
- SthiraRoute plan — windows and curfews kept, 0 unserved



A full day in each city, run headless · nothing staged

WHAT THE FOUR DAYS THREW AT IT

21 rush orders arrived mid-shift

12 orders cancelled after dispatch

4 trucks broke down

20 GPS blackouts

4 municipal closures

17 re-plans priced and approved

WHAT CAME OUT THE OTHER SIDE

69 of 69

orders delivered · 0 freight lost

97% on time across the four days

0 orders silently dropped

Mumbai and Bengaluru finished 100% on time;
Delhi and Hyderabad each lost a single window.

Who feels it



Dispatcher

One board instead of a whiteboard and six
phone calls



Dock & warehouse

Load order from the packer: first drop nearest
the door



Driver

A manifest that matches how the truck is really
loaded



Customer

A promised window the plan is built to keep



Fleet owner & city

No-entry hours obeyed, and an audit trail
behind every call



Economic

Ten to thirty per cent fewer kilometres on the same
orders is fuel and driver-hours saved every shift, and
empty running finally has a number on it.



Environmental

Fewer kilometres means less diesel and less CO₂.
Trucks wait a curfew out at a bay instead of idling at a
cordon, and light days take whole trucks off the road.



Social & regulatory

No-entry hours are obeyed by construction, not by a
driver's judgement at a barricade — so lateness stops
being blamed on the person at the wheel.



Resilience

Four headless days absorbed breakdowns, rush
orders, cancellations, GPS blackouts and closures
without losing a single consignment.

Every claim is traceable

Policy from official sources, method from the optimisation literature, numbers from our own harness on the running system — and where we cannot support a claim, we say we are not making it.

Policy & national context

National Logistics Policy, launched 17 Sep 2022 — logistics cost toward single digits, via ULIP and PM GatiShakti.

pmindia.gov.in

NLP three-year update: ULIP now spans 30+ systems and 160+ crore transactions (Aug 2025).

pib.gov.in · [PRID 2167224](#)

ULIP — the interface joining VAHAN, SARATHI, FASTag, e-Way Bill, ICEGATE and state no-entry layers.

goulip.in

PM GatiShakti — DPIIT portals and connected initiatives.

dpiit.gov.in

Invest India — the four NLP pillars (IDS, ULIP, ELOG, SIG) beside GatiShakti and Bharatmala.

investindia.gov.in

Our four desks map onto roles the policy already names — shipper, carrier, dock, driver — so a pilot needs no new vocabulary.

[our position](#)

The size of the problem

World Bank: India ranks 38 of 139 on the Logistics Performance Index 2023; the goal is top 25 by 2030.

worldbank.org

NCAER / DPIIT: logistics cost assessed at 7.8–8.9% of GDP for 2021–22 under that methodology.

ncaer.org

The older ~13–14% of GDP narrative sits against that assessment because the two methods differ.

pmindia.gov.in · ncaer.org

So our KPIs are deliberately fleet-operational — km, on-time rate, empty running, payload used.

[our position](#)

Every rupee in the objective is a configurable cost model, tunable to an operator's economics.

[our position](#)

Our benchmark is like-for-like on one shift, on one road network — never extrapolated to a fleet-year.

[our position](#)

Optimisation research

Google OR-Tools — vehicle routing framework, distance matrices and constrained variants.

developers.google.com/optimization

OR-Tools routing options — Guided Local Search documented as generally the most efficient metaheuristic here.

developers.google.com

Gmira et al., *Transportation Research Part C*, 2021 — dynamic VRP with parallel ALNS re-optimisation.

sciencedirect.com

Ammouriova et al., *Applied Sciences*, 2022 — simheuristics for VRP under uncertainty.

mdpi.com

Tresca et al., IEEE 2024 — matheuristic delivery planning for the dynamic VRP.

ieeexplore.ieee.org

3L-CVRP — the field's warning that a geometric packer inside every route move is intractable, which is why we pack once per settled route.

[our reading of the field](#)

Engines, maps & market

OSRM vs Valhalla vs GraphHopper — routing engines compared on algorithms and deployment.

mapsi.dev · geofabrik.de

OSM truck-restriction completeness is a known gap for commercial vehicle routing in India.

help.openstreetmap.org

Mappls / MapmyIndia — vehicle-category routing APIs, heavy vehicles included, for Indian roads.

developer.mappls.com

MapmyIndia InTouch and Mappls IoT — telematics platforms already operating in this market.

intouch.mapmyindia.com

Locus and FarEye — incumbents. Our gap is affordable mid-fleet optimisation with stable re-planning.

locus.sh

MapLibre GL with OpenStreetMap tiles — no per-map-load billing, unlike the proprietary map SDKs.

maplibre.org

What we measured ourselves, rather than cited

53 pytest tests pass in 70 seconds — 3D packing and LIFO (16), the routing objective (14), the shift simulator (14), the HTTP API (7) and the assistant (2). Alongside them, **93 interface checks** run in headless Chromium and **45 assertions** run end to end against the live API.

How the numbers on these slides were produced

A single script seeds each city, plans the shift, records the KPIs and the greedy baseline, then runs the whole simulated day and prints the scorecard. Every figure here comes from that run, over live OSRM road distances — re-runnable in about four minutes for all four cities.

What we explicitly do not claim

That a student platform shifts national logistics cost as a share of GDP. That OpenStreetMap matches Mappls for Indian addresses. That we have done a formal prior-art search. The city packs are India-shaped but synthetic, and curfew windows are plausible templates, not gazetted law.